CSE563 — Unit I Numerical Worksheet

Basic Mathematics in Computer Science

40 questions • CO1 • Student Worksheet

Instructions

Attempt all 40 questions. Every question is numerical — show your working, since the method carries most of the marks. Questions are grouped by topic and run roughly in increasing difficulty within each group. Refer to the *Unit I Numerical Guide* for a worked example of each problem type before you begin.

A. Number Theory (Q1–Q10)

Q1. Find the quotient q and remainder r when $a = -73$ is divided by $d = 8$, so that $a = dq + r$ with $0 \le r < d$. State what `-73 % 8` returns in C.

Q2. Compute $\gcd(2166, 6099)$ using the Euclidean algorithm, showing every division step.

Q3. Find $\gcd(414, 662)$ and $\operatorname{lcm}(414, 662)$.

Q4. Use the extended Euclidean algorithm to find integers x, y with $138x + 82y = \gcd(138, 82)$. Verify your answer.

Q5. Find the multiplicative inverse of 9 modulo 31.

Q6. Determine whether $12^{-1} \bmod 18$ exists. Justify your answer.

Q7. Compute $5^{38} \bmod 13$ using Fermat's little theorem, and confirm by square-and-multiply.

Q8. Compute $\phi(1000)$ and $\phi(97)$.

Q9. RSA with $p = 3$, $q = 11$, $e = 7$. Find n, $\phi(n)$ and the private exponent d, then encrypt $M = 5$.

Q10. Determine whether 391 is prime. If composite, give its prime factorisation and the number of positive divisors.

B. Set Theory (Q11–Q15)

Q11. $A = \{p, q, r, s\}$. State $|\mathcal{P}(A)|$, the number of proper subsets, and the number of 2-element subsets.

Q12. $U = \{1, \ldots, 12\}$, $A = \{2, 4, 6, 8, 10, 12\}$, $B = \{3, 6, 9, 12\}$. Compute $A \cap B$, $A \cup B$, $A - B$, $\overline{B}$

and $A \oplus B$.

Q13. In a class of 80, 50 play cricket, 35 play football and 18 play both. How many play (a) at least one, (b) neither, (c) exactly one?

Q14. Of 150 developers: 85 know Python, 70 know Java, 60 know C++, 35 know Python and Java, 30 know Python and C++, 25 know Java and C++, 15 know all three. Find the number who know at least one, none, and exactly two languages.

Q15. $A = \{1, 2\}$, $B = \{x, y, z\}$. Find $|A \times B|$, list three elements of $A \times B$, and state $|\mathcal{P}(A \times B)|$.

C. Boolean Algebra (Q16–Q21)

Q16. Simplify $F = AB + A\overline{B} + \overline{A}B$ using Boolean laws, naming each law.

Q17. Simplify $F = \overline{\overline{A} + \overline{B}} + AB\overline{C}$.

Q18. Write the canonical SOP and POS for $F(A, B, C) = \sum m(0, 3, 5, 6)$.

Q19. Minimise $F(A, B, C) = \sum m(1, 3, 5, 7)$ using a K-map.

Q20. Minimise $F(A, B, C) = \sum m(0, 1, 2, 3, 6, 7)$ using a K-map and write the minimal SOP.

Q21. Minimise $F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$ using a 4-variable K-map.

D. Logic (Q22–Q27)

Q22. Construct the truth table for $(p \vee q) \rightarrow (p \wedge q)$ and classify it.

Q23. Show that $p \to q \equiv \neg q \to \neg p$ using a truth table.

Q24. Prove algebraically (no truth table) that $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$.

Q25. Test the validity of: "If the server is down, users complain. Users are not complaining. Therefore the server is not down." Name the rule used.

Q26. Test the validity of: "If it rains, the match is cancelled. The match was cancelled. Therefore it rained."

Q27. Negate $\exists x \, \forall y \, \big(P(x, y) \wedge Q(y)\big)$ and push the negation as far inward as possible.

E. Calculus (Q28–Q32)

Q28. Differentiate $y = (3x^2 + 1)^4$.

Q29. Differentiate $y = x^2 \ln x$ and find the value of $x > 0$ at which the derivative is zero.

Q30. For $f(x, y) = 3x^2y + 2y^3 - 5x$, find $\partial f/\partial x$, $\partial f/\partial y$ and ∇f at $(1, 2)$.

Q31. Minimise $L(w) = 3w^2 - 18w + 40$ analytically, then perform two gradient-descent steps from $w_0 = 0$ with $\alpha = 0.1$.

Q32. Evaluate $\displaystyle\int_0^2 (3x^2 - 4x + 1)\,dx$.

F. Mathematical Induction (Q33–Q36)

Q33. Prove by induction that $\displaystyle\sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}$ for all $n \geq 1$.

Q34. Prove by induction that $5 \mid (n^5 - n)$ for all integers $n \geq 0$.

Q35. Prove by induction that $n! > 2^n$ for all $n \geq 4$.

Q36. Use strong induction to prove that every integer $n \geq 8$ can be written as a sum of 3s and 5s.

G. Relations and Functions (Q37–Q40)

Q37. On $A = \{1, 2, 3, 4\}$ let $R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (3,4), (4,3)\}$. Determine whether R is an equivalence relation, and if so list its equivalence classes.

Q38. Draw the Hasse diagram of $(D_{24}, |)$ where $D_{24} = \{1, 2, 3, 4, 6, 8, 12, 24\}$. State the least and greatest elements and find $\text{glb}(8, 12)$ and $\text{lub}(8, 12)$.

Q39. For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x + 7$ and $g(x) = x^2$, compute $(f \circ g)(3)$, $(g \circ f)(3)$ and $f^{-1}(x)$.

Q40. Evaluate $\lfloor -4.3 \rfloor$, $\lceil -4.3 \rceil$, $\lfloor \log_2 500 \rfloor$. How many bits are needed to represent the decimal number 500, and how many 512-byte disk blocks are needed for a 5000-byte file?